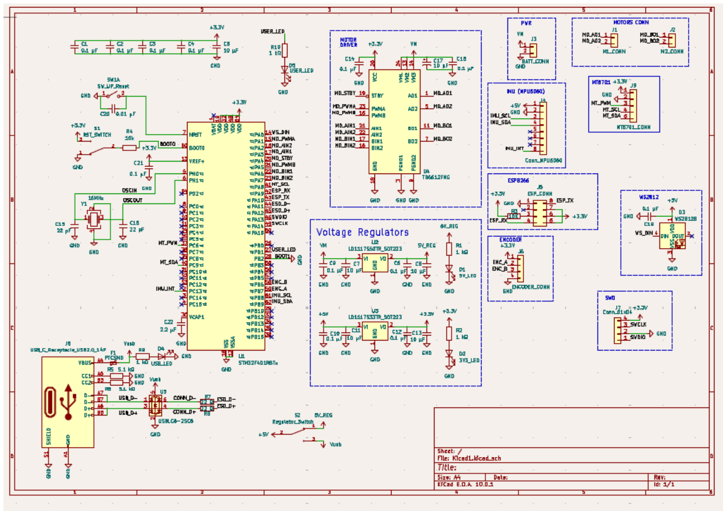
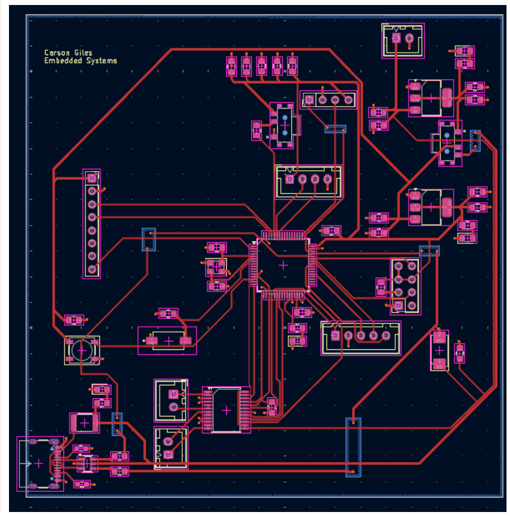
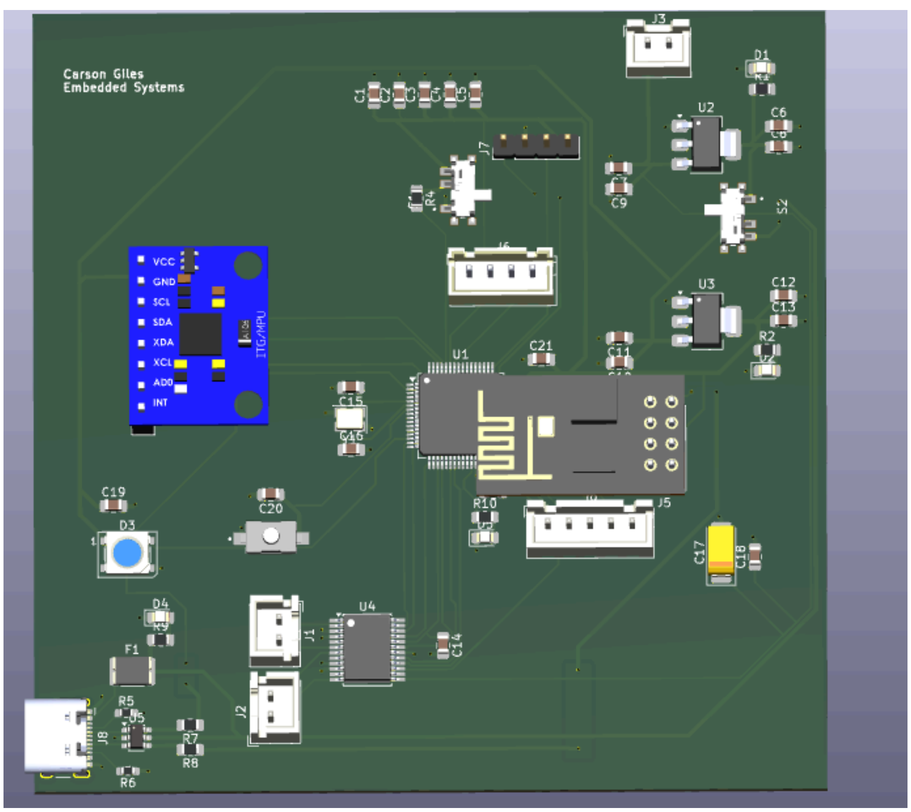
3 wheel balance robot lab report

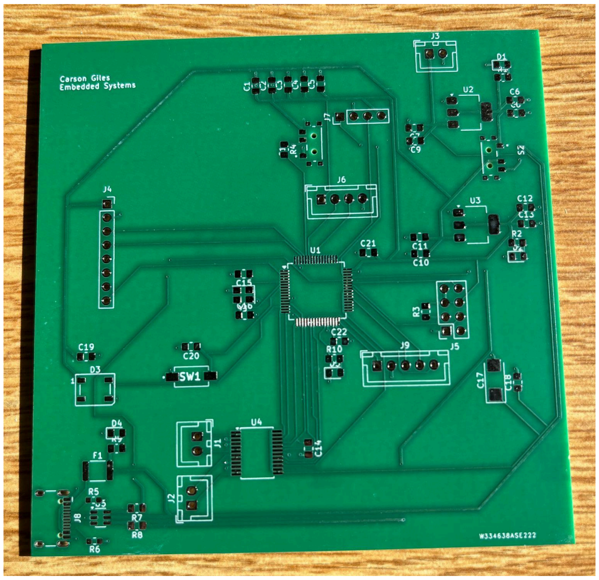
The goal of this project was to design and construct a robot that could balance on 3 wheels from scratch. We were given the wire diagrams and connections. Then we designed the PCB on KiCad. I accidentally made the board really big because I wrote down the wrong number for how long the edges should be. Unfortunately later in the process I found out I messed up part of the differential pair wiring in the design. This caused the usb not to register when connected.

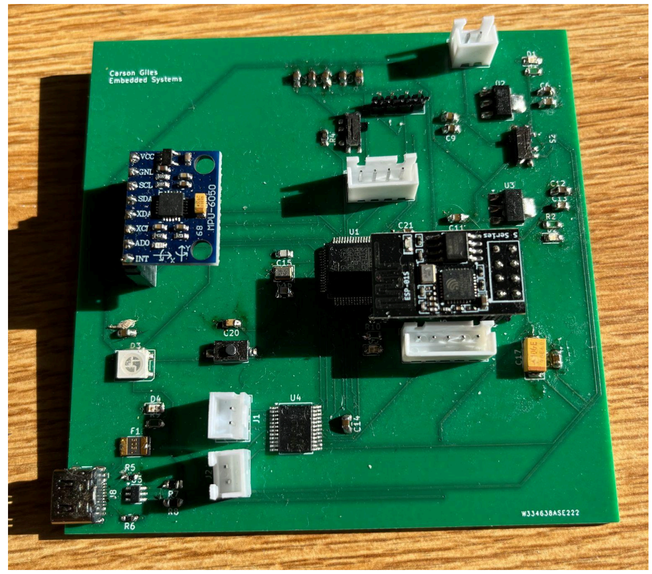




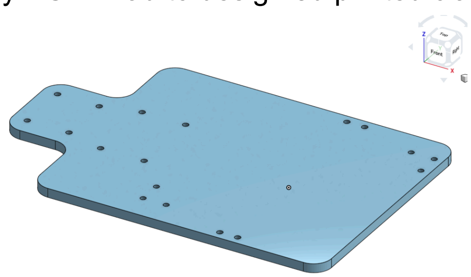


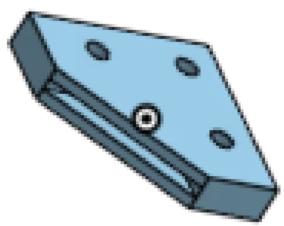
After receiving the board the next part of the project was to solder it. This part went really well and there isn't much to note about it.

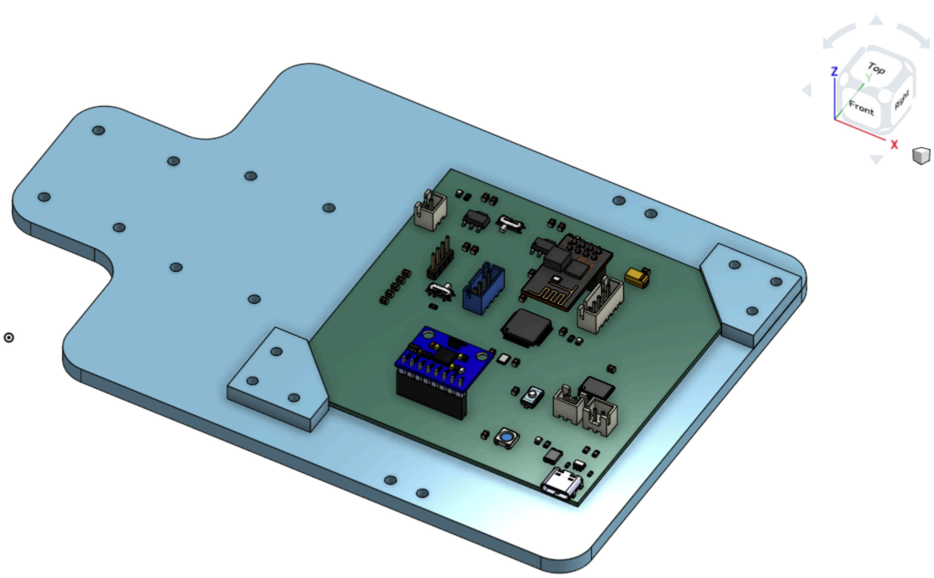




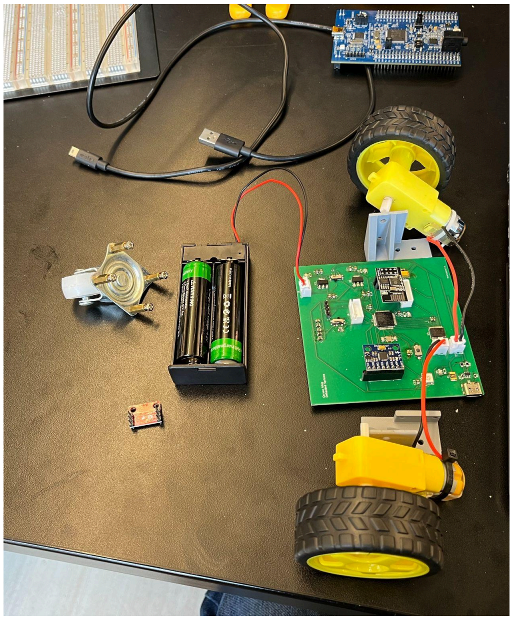
Next I designed the chassis for the vehicle on Onshape. Because I had forgotten to add holes to my PCB I had to design 3d printed clamps that allowed for its attachment.

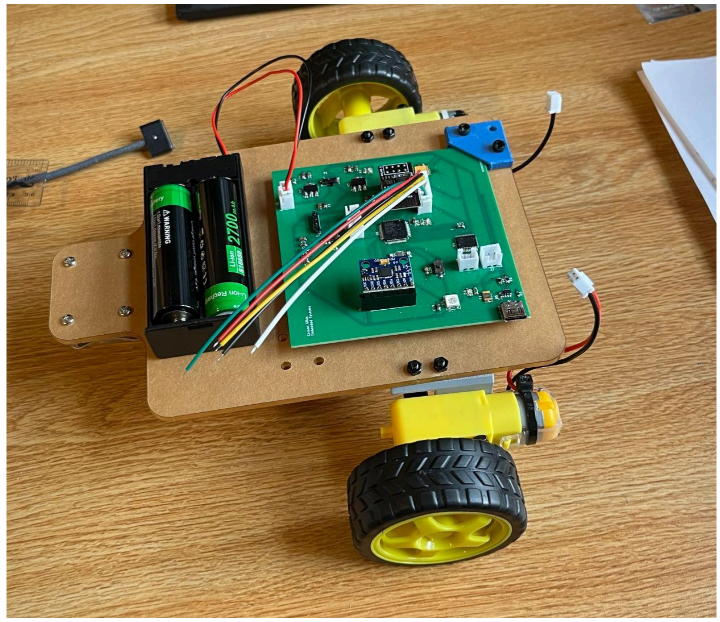






After using the DXF file to cut the chassis out of acrylic and printing out the clamp I designed and assembled the robot. The increased size of the pcb made the chassis much larger then others designs. Additionally a misunderstanding of how the motors attached made it a bit wider then it needed to be. Because of this I was only able to get one made but everything turned out fine. I attached the motors, battery holder, and wheels given to me in class. The size of my chassis also meant the wires were too short to attach to the board.





The robot came together well, but since I messed up the usb port I wasn't able to upload the code. Power connects and the leds turn on but stm32 doesn't recognize the chip. Because of this I wasn't able to get the wifi controller to work.